

EAPO-NET: Adaptive Energy-Aware Profit Optimization for Next-Generation RANs

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Abstract—The large-scale deployment of 5G networks has made energy-efficient operation of radio access networks a critical challenge, requiring the joint optimization of energy consumption, revenue, and quality-of-service compliance under dynamic traffic conditions. In this paper, we propose EAPO-NET, a Multi-Agent Reinforcement Learning (MARL) framework that optimizes these objectives across the Open Radio Access Network (O-RAN) control hierarchy. EAPO-NET employs a heterogeneous multi-agent architecture that aligns learning agents with radio unit (RU) level scheduling, admission control, and node-level energy management functions. Leveraging centralized training with decentralized execution (CTDE) and Value Decomposition Networks, the framework enables fully decentralized inference without inter-agent communication, ensuring scalability and practical deployment. Simulation results show that EAPO-NET learns near-optimal control policies that simultaneously reduce energy consumption, maximize network profit, and satisfy quality of service requirements, while achieving optimization performance comparable to integer linear program-based solutions with significantly greater scalability and adaptability.

Index Terms—O-RAN, energy efficiency, MARL, RU-level scheduling, admission control.

I. INTRODUCTION

The rapid growth of mobile data traffic and heterogeneous service demands has accelerated large-scale 5G deployment, with more than 3.5 million base stations (BSs) already operating worldwide [1]. While densification improves capacity and coverage, it significantly increases energy consumption. In particular, Radio Access Network (RAN) infrastructure accounts for most network power usage, with Radio Units (RUs) contributing up to 80% due to continuous physical-layer operations such as transmission, reception, and signal processing [1], [2]. As 5G evolves toward wider bandwidths and denser deployments, improving RU energy efficiency becomes critical. The disaggregated and virtualized Open Radio Access Network architecture enhances the flexibility and programmability of RAN resource management [3], [4]. By separating hardware and software and introducing entities such as the Near-Real-Time RAN Intelligent Controller (Near-RT RIC) and O-Cloud, O-RAN enables dynamic orchestration of radio and compute resources [4]. Although not originally designed for energy efficiency, its flexibility enables energy-aware RU operation and adaptive DU/CU placement, particularly for heterogeneous services with diverse quality-of-service (QoS) requirements [5].

Empirical studies show that 5G traffic is highly bursty, with millisecond-scale idle intervals [6], [7]. This motivates

fast control mechanisms that exploit short transmission gaps for micro-level energy savings [8], [9]. At larger timescales (seconds to minutes), traffic varies due to mobility and diurnal patterns, enabling structural energy-saving strategies. For instance, lightly loaded RUs can be switched off during off-peak periods, with traffic offloaded to neighboring RUs when coverage allows [10]. Similarly, capacity-layer RUs may be deactivated by handing UEs to macro cells [11]. These observations highlight the need for context-aware decisions that jointly consider traffic, user distribution, and service requirements.

In practice, RU sleep mode¹ is coupled with resource allocation and Virtual Network Function (VNF) placement on the O-Cloud [3], [4]. Joint optimization of these functions increases RU sleep opportunities and improves cloud consolidation, reducing total energy consumption. However, it must also satisfy QoS constraints and operator cost considerations [5]. The O-RAN architecture supports this via xApps in the Near-RT RIC for radio control and dApps on the O-Cloud for VNF placement [4], [12].

In this paper, we propose a joint optimization framework for RU sleep mode, UE admission control, and DU/CU² VNF placement in O-RAN. The objective maximizes net profit defined as UE revenue minus energy cost of active RUs, nodes, and VNFs. This captures the trade-off between admitting more UEs (higher revenue) and reducing energy consumption via sleep modes (lower cost), while satisfying heterogeneous QoS constraints. We propose EAPO-NET, a MARL-based framework aligned with O-RAN, implementing Near-RT RIC xApps for radio control, an admission xApp, and O-Cloud dApps for VNF placement.

EAPO-NET employs a centralized training with decentralized execution (CTDE) architecture from Value Decomposition Networks (VDN) [15] built upon the model-free DQN algorithm. Taking as input a representation of current network conditions, EAPO-NET includes three types of cooperative agents — RU xApp agents, an admission xApp agent, and nodes dApp agents — each maintaining a local policy network that observes only its own domain-specific state and selects actions independently at execution time. During training, each agent type minimizes its own Q-learning loss function using

¹In this work, *sleep mode* switches OFF the entire RU, whereas *blanking* disables a subset of RBs while the RU remains active.

²Co-locating DU/CU functions onto fewer active servers allows unused servers to be switched off, e.g., [13], [14].

targets computed under the CTDE framework, where joint observations and actions are exploited to coordinate learning without requiring inter-agent communication during execution. This decoupling allows EAPO-NET to scale to real O-RAN deployments while retaining the global optimization perspective needed to balance RU energy efficiency and O-Cloud consolidation against UE QoS demands.

A key challenge in designing EAPO-NET is effectively modeling the interactions among RU xApp agents, the admission xApp agent, and the node dApp agents. In our O-RAN system, radio-level energy savings are achieved through RB blanking and RU sleep mode, while compute-level resource consumption is managed by DU/CU placement and O-Cloud node activation decisions. Additionally, the admission agent focuses on maximizing MNO profits. Training these agents in isolation can lead to suboptimal coordination; for instance, putting an RU to sleep eliminates its RB blanking budget, while spinning up an additional O-Cloud node to host a DU may be unnecessary if the corresponding RU remains inactive. This misalignment not only results in inefficient use of resources but also diminishes the potential gains when a UE is admitted to the system. To capture these inter-dependencies, EAPO-NET employs a centralized training phase in which the Q-learning loss functions of the three types of cooperative agents jointly capture their interactions, allowing each agent's updates to account for the joint effect of all five action dimensions: RU ON/OFF state, RB blanking ratio, the admission or rejection of a UE requesting access, O-Cloud node ON/OFF state, and DU/CU placement per active RU. This coordination is achieved without inter-agent communication at execution time, preserving the decentralized structure required for deployment as xApps and dApps in a real O-RAN system. Consequently, this approach optimizes UEs' admission for profit, ensuring that each agent's actions enhance operational efficiency while maximizing the potential gains from admitting users to the network.

This work advances the state-of-the-art as follows:

- 1) We formulate the joint RU sleep mode, UE admittance control, and DU/CU VNF placement in a 5G O-RAN system.
- 2) We model the problem as an MDP and design EAPO-NET, a MADRL-based framework deployed as xApps within the Near-RT RIC and as dApps on O-Cloud nodes, enabling real-time optimization under non-stationary network conditions.
- 3) We evaluate EAPO-NET against a B&C solver, showing near-optimal performance with significantly better scalability in large deployments.

II. BACKGROUND

5G and beyond networks rely on the 5G New Radio (NR) framework, which provides a flexible and scalable frame design capable of supporting a wide range of services, with varying priority levels and distinct QoS requirements [5]. The 5G NR frame structure is organized around a 10 ms radio frame that is partitioned into subframes of 1 ms duration. Each

subframe is then divided into slots, where a slot may contain up to 14 symbols [16], [17]. A key feature of this architecture is its adaptable numerology, which defines the granularity of the time-frequency resource grid through different subcarrier spacings. As wireless systems progress toward 6G, it is expected that the existing frame structure and numerology concepts will be preserved to maintain compatibility with earlier generations.

O-RAN adopts an open and modular architecture that enables interoperability among multi-vendor equipment while improving network management and performance [3]. Its main components are the RU, DU, and CU. The RU handles lower-layer radio functions, including PHY processing and RF operations at cell sites. The DU processes lower protocol layers, mainly L2 and part of L3, including MAC, RLC, and PDCP functions, while also supporting scheduling, beamforming, and radio resource management [3], [4]. The CU manages upper-layer protocols such as SDAP, RRC, and NAS. O-RAN also integrates the Near-RT RIC, which provides near-real-time control, monitoring, and optimization by interacting with the DU and CU [4]. It hosts xApps for functions such as traffic steering, resource allocation, and energy optimization, while dApps can provide distributed intelligence and decentralized decision-making across the network [4].

Energy saving in 3GPP has become increasingly important due to dense BS deployments and fluctuating traffic demand [18]. A widely studied approach is the ON/OFF sleep mechanism for RUs/BSs, where underutilized nodes enter low-power states during low-traffic periods [10], [11]. These mechanisms may be static or dynamic and often require coordination to avoid coverage gaps or service degradation [10]. In O-RAN, centralized intelligence such as the Near-RT RIC or distributed learning agents can determine RU/BS activation states while considering mobility, QoS, and energy costs. Similarly, O-Cloud energy-saving strategies consolidate DU/CU VNFs onto fewer computing nodes, allowing lightly loaded servers to enter low-power states while preserving service and latency requirements [14], [19], [20]. Complementing node-level energy savings, *RB blanking* operates at the resource-allocation level [9]. Certain RBs are intentionally muted to reduce interference and transmission power, particularly under low-traffic conditions. By selectively blanking RBs, networks can lower power amplifier consumption and inter-cell interference, especially in dense deployments [21]. Combined with sleep modes, RB blanking provides a finer-grained energy-saving mechanism by partially deactivating radio resources without shutting down network elements [8], [9]. Together, these techniques support energy-efficient 5G and beyond networks while maintaining performance.

III. PROBLEM DEFINITION

We first present the system model and its assumptions, and then formulate the problem as a stochastic optimization problem.

A. System Model

The network infrastructure consists of a set $\mathcal{R}$ of RUs, each composed of a set $\mathcal{B}$ of RBs and serving a set $\mathcal{U}$ of UEs over the same geographical area. The computing infrastructure is modeled by a set $\mathcal{V}$ of cloud nodes hosting 5G VNFs, namely DU and CU functions. Each RU must be associated with exactly one DU and one CU, while cloud nodes may host multiple VNFs subject to resource availability. We assume a fully connected topology between RUs and cloud nodes.

Moreover, any network component (RU or cloud node) can be switched to sleep mode, as discussed in Section II. We use variables

$$\mathbf{x} \in \{0, 1\}^{\mathcal{R} \cup \mathcal{V}}, \quad (1)$$

to indicate that a component $i \in \mathcal{R} \cup \mathcal{V}$ is in active mode, i.e., $x_i = 1$, or in sleep mode, i.e., $x_i = 0$.

1) *UE Admission and SLA*: UEs may either be admitted into the system or rejected. Admitted UEs must be associated with exactly one RU, while rejected UEs receive no service. The admission and association decisions are controlled through variables

$$\mathbf{y} \in \{0, 1\}^{\mathcal{R} \times \mathcal{U}}, \quad (2)$$

where $y_{r,u} = 1$ if UE u is associated with RU r and $y_{r,u} = 0$, otherwise. Hereafter, we refer to $\mathbf{y}$ as the *admission plan*. In 5G systems, SLAs define the QoS guarantees associated with each UE and directly impact the MNO's economic outcomes. In this work, SLA requirements are modeled solely through downlink throughput demands. Therefore, if UE $u \in \mathcal{U}$ is admitted, then

- its *individual profit* $p_u \in \mathbb{R}_{\geq 0}$ is generated for the MNO in line with the SLA, and
- the *number of RBs* $\xi = \{\xi_{r,u} \in \mathbb{N}_{>0} : r \in \mathcal{R}, u \in \mathcal{U}\}$ is consumed from RU r to satisfy its QoS requirements. Parameter $\xi_{r,u}$ changes with the channel conditions³.

2) *RB States*: In this paper, we assume that RUs are equipped with an RB blanking mechanism, as detailed in Section II. When blanking is used, each RB may be (i) available, if it may be scheduled for transmission during the current frame, or (ii) blanked. The available RBs may be in the following states:

- An RB is in *transmitting* state if it is effectively scheduled for data transmission in the current frame. The number of transmitting RBs in RU r is given by

$$g_r^{\text{Tx}}(\mathbf{y}) = \sum_{u \in \mathcal{U}} \xi_{r,u} \cdot y_{r,u}, \quad (3)$$

which depends on the admission plan $\mathbf{y}$.

- An RB is in an *idle* state if it is available but not transmitting. The number of idle RBs in RU r is

$$g_r^{\text{Id}}(\mathbf{y}) = g_r^{\text{Av}}(\mathbf{y}) - g_r^{\text{Tx}}(\mathbf{y}), \quad (4)$$

³We consider that $\xi_{r,u} = \left\lceil \frac{\lambda_u}{\eta_{r,u}} \right\rceil \in \mathbb{N}$ where $\lambda_u \in \mathbb{R}_+$ (in bps) is the average throughput and $\eta_{r,u} \in \mathbb{R}_{\geq 0}$ (in bps/RB) is the achievable wireless link capacity per RB between UE u and RU r .

where

$$g_r^{\text{Av}}(\mathbf{y}) = \begin{cases} g_r^{\text{Tx}}(\mathbf{y}), & \text{if blanking is adopted,} \\ B, & \text{otherwise.} \end{cases} \quad (5)$$

In other words, if blanking is adopted, then $g_r^{\text{Id}}(\mathbf{y}) = 0$, i.e., there are no idle RBs.

3) *VNF Placement*: Each cloud node can implement multiple VNFs, and we introduce DU/CU VNF placement variables

$$\mathbf{z}^{\text{VNF}} \in \{0, 1\}^{\mathcal{R} \times \mathcal{V}}, \text{ VNF} \in \{\text{DU}, \text{CU}\}, \quad (6)$$

to decide whether RU r 's VNF is placed on node i . We denote the set of all VNF placements by $\mathbf{z} = (\mathbf{z}^{\text{DU}}, \mathbf{z}^{\text{CU}})$. The placement of DU and CU functions comes at a computational cost for the cloud nodes. Following a similar approach to prior work (e.g., [22]), we assume that the computing power required to run DU and CU tasks is proportional to the number of transmitting RBs in their associated RU. Therefore, the total computational load (in GOPS) performed by node $i \in \mathcal{V}$ is

$$w_i(\mathbf{y}, \mathbf{z}) = \sum_{r \in \mathcal{R}} g_r^{\text{Tx}}(\mathbf{y}) \cdot (\omega^{\text{DU}} \cdot z_{r,i}^{\text{DU}} + \omega^{\text{CU}} \cdot z_{r,i}^{\text{CU}}), \quad (7)$$

where $g_r^{\text{Tx}}(\cdot)$ is defined in (3) and parameters ω^{DU} and ω^{CU} are the volumes of computing workload per RB (in GOPS/RB) required for running DU and CU tasks, respectively.

4) *Energy-Aware Profit Model*: In this paper, we consider the energy expense to run network operations at the RU-level and at the cloud-level. The energy consumed by RU r is

$$e_r^{\text{RU}}(\mathbf{x}, \mathbf{y}) = [\epsilon^{\text{Av}} \cdot g_r^{\text{Av}}(\mathbf{y}) + \epsilon^{\text{Tx}} \cdot g_r^{\text{Tx}}(\mathbf{y})] \cdot x_r, \quad (8)$$

where ϵ^{Av} and ϵ^{Tx} (both in kWh/RB) are the power consumed to keep one RB in the available state and in the transmitting state [21]. The energy consumed by cloud node i is

$$e_i^{\text{Cloud}}(\mathbf{x}, \mathbf{y}, \mathbf{z}) = (\epsilon_i^{\text{C}} \cdot w_i(\mathbf{y}, \mathbf{z}) + \epsilon_i^{\text{Base}}) \cdot x_i, \quad (9)$$

where ϵ_i^{C} denotes the energy consumed to perform a unit of computational tasks (in kWh/GOPS) and ϵ_i^{Base} represents the baseline power required to keep a cloud node active. The total *energy expense* to run network operations is

$$E(\mathbf{x}, \mathbf{y}, \mathbf{z}) = \sum_{r \in \mathcal{R}} e_r^{\text{RU}}(\mathbf{x}, \mathbf{y}) + \sum_{i \in \mathcal{V}} e_i^{\text{Cloud}}(\mathbf{x}, \mathbf{y}, \mathbf{z}), \quad (10)$$

where $e_r^{\text{RU}}(\cdot)$ and $e_i^{\text{Cloud}}(\cdot)$ are introduced in (8) and (9), respectively.

The *revenue* obtained from admitting UEs into the network corresponds to the total income generated by the admitted UEs, i.e.,

$$H(\mathbf{y}) = \sum_{r \in \mathcal{R}} \sum_{u \in \mathcal{U}} p_u \cdot y_{r,u}, \quad (11)$$

where p_u denotes the revenue associated with admitting UE u , which may vary across UEs according to business policies or service priorities.

B. Problem Formulation

Our goal is to maximize the system profit, defined as the difference between the revenue and energy consumption, i.e.,

$$P(\mathbf{x}, \mathbf{y}, \mathbf{z}) = H(\mathbf{y}) - \alpha \cdot E(\mathbf{x}, \mathbf{y}, \mathbf{z}) \quad (12)$$

where $H(\cdot)$ is given by (11), $E(\cdot)$ is given by (10), and α the energy-cost weight (in \$/kWh).

Intuitively, admitting more UEs increases the MNO's revenue, as each admitted UE generates a profit depending on its service characteristics and associated priority. However, serving these UEs requires activating radio and computing resources, thereby increasing the network energy consumption. There is a trade-off: the revenue brought by a UE justifies the additional operational cost required to serve it.

The maximization of (12) is subject to several constraints, which are described next.

C1: At any time, each active RU must be associated with exactly one VNF of each type (i.e., one DU and one CU) deployed on cloud nodes, i.e.,

$$\sum_{i \in \mathcal{V}} z_{r,i}^v = x_r, \forall r \in \mathcal{R}, v \in \{\text{DU}, \text{CU}\}. \quad (13)$$

C2: Any admitted UE can only be associated with an active RU and to one RU only, i.e.,

$$y_{r,u} \leq x_r, \forall r \in \mathcal{R}, \forall u \in \mathcal{U}. \quad (14)$$

$$\sum_{r \in \mathcal{R}} y_{r,u} \leq 1, \forall u \in \mathcal{U}. \quad (15)$$

C3: The set of admitted UEs in each RU is bounded by the existing number of RBs, i.e.,

$$g_r^{\text{Tx}}(\mathbf{y}) \leq B, \forall r \in \mathcal{R}, \quad (16)$$

where $g_r^{\text{Tx}}(\cdot)$ is introduced in (3).

C4: The number of VNFs hosted by cloud node i is limited by its processing capacity μ_i , i.e.,

$$w_i(\mathbf{y}, \mathbf{z}) \leq \mu_i, \quad \forall i \in \mathcal{V}, \quad (17)$$

where we introduce function $w_i(\cdot)$ in (7).

Problem 1. The problem is to maximize the total profit (12) while satisfying system constraints C1–C4.

$$\begin{aligned} & \underset{\mathbf{x}, \mathbf{y}, \mathbf{z}}{\text{maximize}} && P(\mathbf{x}, \mathbf{y}, \mathbf{z}) \\ & \text{subject to} && (1), (2), (6), (13) - (17) \end{aligned}$$

Proposition 1. Problem 1 is $\mathcal{NP}$ -Hard.

Proof. Consider the decision version of Problem 1 defined as follows:

Problem 2 (DEC-P1). Given an instance of Problem 1, i.e., an assignment of parameters

$$(\mathcal{R}, \mathcal{U}, \mathcal{V}, B, \xi, p, \omega^{DU}, \omega^{CU}, \epsilon^{Av}, \epsilon^{Tx}, \epsilon^C, \epsilon^{Base}, \mu, \sigma)$$

and a target value $K \in \mathbb{R}$, does there exist a feasible assignment $(\mathbf{x}, \mathbf{y}, \mathbf{z})$ satisfying constraints (13)–(17) such that $P(\mathbf{x}, \mathbf{y}, \mathbf{z}) \geq K$?

We first show that DEC-P1 $\in \mathcal{NP}$. For any given candidate solution $(\hat{\mathbf{x}}, \hat{\mathbf{y}}, \hat{\mathbf{z}})$, verifying all constraints and evaluating $P(\hat{\mathbf{x}}, \hat{\mathbf{y}}, \hat{\mathbf{z}})$ each take time polynomial in $|\mathcal{R}|, |\mathcal{U}|, |\mathcal{V}|$ as equations (3)–(11) are sums over $\mathcal{R}, \mathcal{U}, \mathcal{V}$ with no combinatorial subroutines. In other words, nothing in this verification step requires solving another hard search/optimization problem. Hence, a solution can be verified in polynomial time, so DEC-P1 $\in \mathcal{NP}$.

The next step is to reduce DEC-P1 from another well-established $\mathcal{NP}$ -Complete problem. For that, we use the 0/1 Knapsack decision problem that we present as follows:

Problem 3 (KNAPSACK). Given n items with weights $w_i \in \mathbb{N}$ and values $v_i \in \mathbb{N}$ for $i = 1, \dots, n$, a capacity $W \in \mathbb{N}$, and a target $K' \in \mathbb{N}$, does there exist $S \subseteq \{1, \dots, n\}$ with $\sum_{i \in S} w_i \leq W$ and $\sum_{i \in S} v_i \geq K'$?

The KNAPSACK problem is $\mathcal{NP}$ -Complete [?].

Given an arbitrary KNAPSACK instance, we construct the following instance of DEC-P1 in polynomial time:

- $\mathcal{R} = \{1\}$ (a single RU), $\mathcal{U} = \{1, \dots, n\}$ (one UE per item), $\mathcal{V} = \{1\}$ (a single cloud node);
- $B = W$, and $\xi_{1,u} = w_u$ for each $u \in \mathcal{U}$;
- $p_u = v_u$ for each $u \in \mathcal{U}$;
- $\omega^{DU} = \omega^{CU} = 0$, so by (7), $w_1(y, z) = 0$ for every y, z ; set $\mu_1 \geq 0$ arbitrarily, so C4 (17) is satisfied unconditionally;
- $\epsilon^{Av} = \epsilon^{Tx} = \epsilon_1^C = \epsilon_1^{Base} = 0$, so by (8)–(10), $E(\mathbf{x}, \mathbf{y}, \mathbf{z}) = 0$ for every $\mathbf{x}, \mathbf{y}, \mathbf{z}$, and hence $P(\mathbf{x}, \mathbf{y}, \mathbf{z}) = H(y)$ regardless of σ ;
- target $K = K'$.

This transformation is linear in the size of the KNAPSACK instance (n items map to n UEs, plus a constant number of RUs, nodes, and parameters), so it runs in polynomial time.

Finally, we show the KNAPSACK instance is a “yes” instance if, and only if, the constructed DEC-P1 instance is a “yes” instance. Since $\mathcal{V} = \{1\}$, C1 (13) forces $z_{1,1}^{DU} = z_{1,1}^{CU} = x_1$, which is always satisfiable and costless (as $\omega^{DU} = \omega^{CU} = 0$). Since $E \equiv 0$, no assignment is penalized in the objective through energy, so without loss of generality the optimal solution sets $x_1 = 1$: if $x_1 = 0$, then C2 (14) forces $y_{1,u} = 0$ for all u , giving $H(y) = 0$, which is dominated by any solution with $x_1 = 1$ whenever $K > 0$ (and both cases coincide trivially when $K \leq 0$, since the empty knapsack is always feasible). With $x_1 = 1$, C2 is automatically satisfied for every y , so the only active constraint is C3 (16):

$$\begin{aligned} g_1^{\text{Tx}}(\mathbf{y}) &= \sum_{u \in \mathcal{U}} \xi_{1,u} y_{1,u} \\ &= \sum_{u \in \mathcal{U}} w_u y_{1,u} \leq B = W. \end{aligned}$$

The objective reduces to

$$\begin{aligned} P(\mathbf{x}, \mathbf{y}, \mathbf{z}) &= H(\mathbf{y}) = \sum_{u \in \mathcal{U}} p_u y_{1,u} \\ &= \sum_{u \in \mathcal{U}} v_u y_{1,u}. \end{aligned}$$

Thus, maximizing $P(\cdot)$ subject to the constructed instance's constraints is exactly the problem of choosing $S = \{u : y_{1,u} = 1\} \subseteq \{1, \dots, n\}$ maximizing $\sum_{u \in S} v_u$ subject to $\sum_{u \in S} w_u \leq W$. In other words, it is exactly the 0/1 Knapsack optimization problem on the original instance. Hence there exists a feasible $(\mathbf{x}, \mathbf{y}, \mathbf{z})$ with $P(\mathbf{x}, \mathbf{y}, \mathbf{z}) \geq K$ if, and only if, there exists $S \subseteq \{1, \dots, n\}$ with $\sum_{i \in S} w_i \leq W$ and $\sum_{i \in S} v_i \geq K'$.

Since KNAPSACK is $\mathcal{NP}$ -Complete and reduces to DEC-P1 in polynomial time, combined with DEC-P1 $\in \mathcal{NP}$, DEC-P1 is $\mathcal{NP}$ -Complete. As Problem 1 is the optimization version of DEC-P1 (i.e., an oracle for Problem 1 solves DEC-P1 by comparing the optimum to K), Problem 1 is $\mathcal{NP}$ -Hard. $\square$

In this Problem 1, we have three coupled decisions: (i) network component activation, (ii) UE admission control, and (iii) placement of DU and CU VNFs across the cloud infrastructure. By coordinating these decisions, the system seeks to *consolidate* traffic on a limited subset of infrastructure resources whenever possible, allowing unused RUs or cloud nodes to remain in low-power states. However, Problem 1 faces two major challenges:

- The presence of ξ in the system dynamics makes it difficult to derive deterministic policies that remain effective across all possible realizations.
- In practice, the different decision types are handled by distinct system components that cannot necessarily communicate and coordinate their actions toward a globally optimal solution.

Multi-Agent Reinforcement Learning (MARL) provides a natural framework to address these challenges. First, by learning through repeated interactions with the environment, MARL agents can implicitly adapt to the stochastic dynamics induced by ξ and optimize the expected long-term objective without requiring explicit analytical models. Second, by assigning an autonomous agent to each decision-making component, MARL naturally supports decentralized execution, allowing agents to learn coordinated behaviors while operating with limited or no real-time communication.

IV. EAPO-NET

In this section, we introduce Adaptive Energy-Aware Profit Optimization in Next-Generation RANs (EAPO-NET), a novel MARL framework for tackling Problem 1. The idea is to decompose the global optimization problem into a set of coordinated local decision processes, each handled by a different agent. Each agent performs localized decisions based on partial system observations, while collectively learning a policy that approximates a globally efficient solution. We first explain EAPO-NET's components, then introduce the underlying Markov Decision Processes (MDPs). Finally, we describe the multi-agent algorithm, its properties, and its distribution across the different components.

A. EAPO-NET Components and MDP Formulation

EAPO-NET is a framework where the multiple agents are distributed among O-RAN's (i) Near-RT RIC, in the form of

xApps, and (ii) O-Cloud nodes, in the form of dApps. In the following, we define the different types of agents and describe how they relate to Problem 1.

1) *Energy Control (EC)*: Each EC agent is deployed as an xApp that controls a single RU's "sleep mode" and "blanking." We introduce the state space of RU r 's EC agent as $\mathcal{O}_r^{\text{EC}} = [x_r, g_r, \beta_r, \mathbf{y}_r]$, where:

- $x_r \in \{0, 1\}$ indicates if the RU is in active mode,
- $g_r \in \mathbb{Z}_{\geq 0}$ is the amount of transmitting RBs,
- $\beta_r \in [0, 1]$ is the fraction of blanked RBs, and
- $\mathbf{y}_r \subseteq \mathcal{U}$ is the list of UEs associated with RU r .

The size complexity of the set of states of EC agents is given by $O(4 \cdot |\mathcal{R}|)$. This formulation is elaborated upon in Section III-A.

Moreover, we introduce the action space of RU r 's EC agent as $\mathcal{A}_r^{\text{EC}} = [a_r, b_r]$, where:

- $a_r \in \{\text{ON}, \text{OFF}\}$ denotes the activation action, such that $a_r = \text{ON}$ sets $x_r = 1$ and vice-versa.
- $b_r \in \{0, 0.25, 0.5, 0.75, 1\}$ denotes the blanking action drawn from a discrete set of percentages.

The action set has a size complexity of EC agents is $O(10 \cdot |\mathcal{R}|)$.

The Lagrangian G^{EC} coming from the EC part is:

$$G^{\text{EC}}(\mathbf{x}, \mathbf{y}) = \sum_{r \in \mathcal{R}} e_r^{\text{RU}}(\mathbf{x}, \mathbf{y}) + \alpha_r^{\text{C3}} [g_r^{\text{Tx}}(\mathbf{y}) - B] \quad (18)$$

2) *VNF Control (VC)*: Each VC agent is deployed as a dApp in a given O-Cloud node that controls which VNFs are locally deployed and which ones should be delegated to other O-Cloud nodes. We introduce the state space of node i 's VC agent as $\mathcal{O}_i^{\text{VC}} = [x_i, z_i, w_i]$, where:

- $x_i \in \{0, 1\}$ indicates if the node is in active or sleep mode,
- $z_i \in \{0, 1\}$ indicates which VNFs are being locally hosted, and
- $w_i \in \mathbb{R}_{\geq 0}$ is the amount of resources consumed, analogous to (7).

The size complexity of the set of states of VC agents is given by $O(3 \cdot |\mathcal{V}|)$. This formulation is elaborated upon in Section III-A. Moreover, we introduce the action space of node i 's VC agent as $\mathcal{A}_i^{\text{VC}} = [a_i, \mathbf{m}_i^{\text{VNF}}]$, where:

- $a_i \in \{\text{ON}, \text{OFF}\}$ denotes the activation action, such that $a_i = \text{ON}$ sets $x_i = 1$ and vice-versa.
- $\mathbf{m}_i^{\text{VNF}} \in \{0, 1\}^{R \times V}$ denotes the migration action for each VNF. Here, such that $m_{i,r,j}^{\text{VNF}} = 1$ indicates that node i is moving RU r 's VNF to node j .

The action set has a size complexity of $O(2 + |\mathcal{R}| \cdot |\mathcal{V}|)$ for VC agents. The Lagrangian G^{VC} coming from the VC part is:

$$G^{\text{VC}}(\mathbf{y}, \mathbf{z}) = \sum_{i \in \mathcal{V}} e_i^{\text{Cloud}}(\mathbf{y}, \mathbf{z}) + \alpha_i^{\text{C4}} [w_i(\mathbf{y}, \mathbf{z}) - \mu_i] \quad (19)$$

3) *Admission Control (AC)*: The AC agent is deployed as an xApp that controls which UEs are admitted to the system and which RU they are associated with. We introduce the state space of the AC agent as $\mathcal{O}^{\text{AC}} = \{\{\mathcal{O}_r^{\text{EC}} : r \in \mathcal{R}\}, \{\mathcal{O}_i^{\text{VC}} : i \in$

$\mathcal{V}\}$, $\xi]$, which considers the states of the other agents and the vector of RBs $\xi = \{\xi_{r,u} : r \in \mathcal{R}, u \in \mathcal{U}\}$ required by the UEs. The size complexity of the set of states of the AC agent is given by $O(|\mathcal{R}| \cdot (4 + |\mathcal{U}|) + 3 \cdot |\mathcal{V}|)$. Moreover, we introduce the action space of the AC agent as $\mathcal{A}^{\text{AC}} = \{0, 1\}^{\mathcal{U} \times \mathcal{R}}$, such that each action is a different UE-to-RU association mapping. The action set has a size complexity of $O(|\mathcal{R}| \cdot |\mathcal{V}|)$ for the AC agent. The Lagrangian G^{AC} coming from the AC part is:

$$G^{\text{AC}}(\mathbf{x}, \mathbf{y}) = \sum_{r \in \mathcal{R}} \sum_{u \in \mathcal{U}} p_u \cdot y_{r,u} + \alpha_r^{\text{C2}} [y_{r,u} - x_r] + \alpha_r^{\text{C2}'} [y_{r,u} - 1] \quad (20)$$

Hence, the total Lagrangian of the system is given by:

$$\begin{aligned} G(\mathbf{x}, \mathbf{y}, \mathbf{z}) &= G^{\text{EC}} + G^{\text{VC}} + G^{\text{AC}} \\ &= P(\mathbf{x}, \mathbf{y}, \mathbf{z}) + \sum_{r \in \mathcal{R}} \alpha_r^{\text{C3}} [g_r^{\text{Tx}}(\mathbf{y}) - B] \\ &\quad + \sum_{i \in \mathcal{V}} \alpha_i^{\text{C4}} [w_i(\mathbf{y}, \mathbf{z}) - \mu_i] \\ &\quad + \sum_{r \in \mathcal{R}} \sum_{u \in \mathcal{U}} \alpha_r^{\text{C2}} [y_{r,u} - x_r] + \alpha_r^{\text{C2}'} [y_{r,u} - 1] \end{aligned} \quad (21)$$

The system can be formalized as a single MDP integrating EC, VC, and AC components containing global state and action spaces defined as

$$\begin{aligned} \mathcal{O} &= \left(\prod_{r \in \mathcal{R}} \mathcal{O}_r^{\text{EC}} \right) \times \left(\prod_{i \in \mathcal{V}} \mathcal{O}_i^{\text{VC}} \right) \times \mathcal{O}^{\text{AC}}, \text{ and} \\ \mathcal{A} &= \left(\prod_{r \in \mathcal{R}} \mathcal{A}_r^{\text{EC}} \right) \times \left(\prod_{i \in \mathcal{V}} \mathcal{A}_i^{\text{VC}} \right) \times \mathcal{A}^{\text{AC}}, \text{ respectively.} \end{aligned}$$

The MDPs' transition dynamics are unknown and cannot be accurately modeled, making it infeasible to derive or exploit transition probabilities. Consequently, model-based solution methods are not applicable, and the problem must be addressed using model-free learning techniques. Furthermore, the MDPs exhibit very large state-action spaces and significant combinatorial complexity, as discussed in detail in our supplementary report [?]. These characteristics render conventional RL approaches impractical and motivate the adoption of DRL methods.

B. EAPO-NET's Algorithm Description

In a multi-agent setting, a fundamental challenge arises from the non-stationarity of the environment: as each agent updates its policy, the environment appears to change from the perspective of every other agent, breaking the Markov assumption and destabilizing independent learning. Furthermore, while inter-agent coordination is desirable, real-time synchronization at every decision step is impractical given the heterogeneous control loop latencies across agents.

To address these challenges, we draw inspiration from the Value Decomposition Networks (VDN) in the CTDE method [15], which resolves the coordination problem by performing (i) *centralized training*, where the agents' policies are obtained by training their underlying DQNs using a global

reward signal and (ii) *decentralized execution*, where the learned policy is applied, requiring no inter-agent message exchange during operation.

Centralized Training Phase: A centralized trainer has access to the joint observation tuple $\langle o_1^t, \dots, o_K^t \rangle$ and maintains one independent utility network $Q_k(o_k, a_k; \theta_k)$ per agent. All agents share a global replay buffer $\mathcal{D}$ that stores transitions $(\mathcal{O}_k(t), \mathcal{A}_k(t), R, \mathcal{O}_k(t+1))$ collected across all k agents. We define the global reward as

$$R_t = G(\mathbf{a}_t(\mathbf{s}_t)) - G(\mathbf{s}_t), \quad (22)$$

where $\mathbf{s}_t \in \mathcal{O}$ is the state at time t , $\mathbf{s}_{t+1} = \mathbf{a}_t(\mathbf{s}_t)$ is the state at time $t+1$ after applying action $\mathbf{a}_t$, and $G(\cdot)$ is the Lagrangian function of Problem 1.

The global reward R_t serves as a common training signal that aligns all agents toward the system-wide objective. Despite sharing the same reward, each agent k learns an independent action-value function by minimizing its own temporal-difference (TD) loss. The target Q-value and loss function are defined as: $y_{\text{total}} = R_t + \gamma \sum_{k=1}^K \max_{a'_k} Q_k(o_k(t+1), a'_k; \theta_k^-)$

$$\text{and } \mathcal{L}(\theta) = \mathbb{E}_{\mathcal{D}} \left[(y_{\text{total}} - \sum_{k=1}^K Q_k(o_k(t), a_k(t); \theta_k))^2 \right]$$

where θ_k^- denotes the parameters of the periodically synchronized target Q-value and γ is the discount factor.

Decentralized Execution Phase: After centralized training, each agent k receives only its trained network θ_k and operates in a fully decentralized manner. At each time slot t , agent k observes solely its local observation $o_k(t)$ and selects its action greedily, without requiring any knowledge of other agents' observations or actions. No inter-agent communication takes place at decision time, and the replay buffer and centralized loss are no longer involved. The IGM condition guarantees that this greedy per-agent maximization remains consistent with the joint optimum found during centralized training.

At execution time, although potential conflicts may arise between agents, they are structurally prevented rather than statistically avoided: the causal ordering enforced within each decision step constrains each agent to act only within the feasible space left by preceding agents, making inter-agent conflicts impossible by construction without requiring any explicit communication.

V. PERFORMANCE EVALUATION

A. Experimental Setup

To evaluate the EAPO-NET framework, we developed a Python-based O-RAN simulation environment instantiated in two configurations: a *small setup* targeting a dense urban micro-cell scenario and a *big setup* representing a larger-scale deployment, with all other system parameters held constant across both configurations.

To implement the MA-DQN learning framework of EAPO-NET, we used PyTorch [23] for scalable, efficient neural network training, enabling practical policy learning. The benchmark optimal solution is obtained by implementing the corresponding ILP described in Problem 1 in Pyomo and

solving it with the GLPK solver [24], which relies on a B&C algorithm, to explore the feasible space and guarantee global optimality systematically. We summarize the system's parameters in Table I.

Setup-Dependent Parameters		
Parameter	Small setup	Big setup
Number of RUs $ \mathcal{R} $	4	15
UE population $ \mathcal{U} $ [min, max]	[10, 65]	[50, 150]
UE data rate λ_u [min, max] Mbps	[1.5, 8.25]	[6, 12]

Fixed System Parameters	
Parameter	Value
Number of O-Cloud nodes $ \mathcal{V} $	7
RU capacity B	100 (RBs)
Achievable wireless capacity $\eta_{u,r}$	0.25 (Mbps/RB)
RU Tx energy ϵ^{Tx}	6 (Wh/RB)
O-Cloud Energy $\epsilon_i^{\text{C}}, \epsilon_i^{\text{Base}}$	100, 30 (Wh)
O-Cloud node computing capacity μ_i^{Comp}	300 (GOPS)

Learning Parameters	
Parameter	Value
Epoch	10000
Activation function	ReLU
Optimizer	Adam
Learning rate	$l_r = 1 \times 10^{-4}$
$\epsilon_0, \epsilon_{\min}, \epsilon_{\text{decay}}$	1.0, 0.01, 0.995
Discount factor	0.91
Replay buffer, Batch size	10^6 , 128

TABLE I: Simulation Parameters

B. Convergence Behavior

We evaluate the convergence of EAPO-NET using three metrics: average episode reward, training loss, and energy consumption. Results for both RB-blanking and non-blanking configurations are shown in Fig. 1.

Fig. 1 shows that both variants converge after approximately 2,000 training episodes. The reward stabilizes, the training loss decreases steadily, and the energy consumption converges, with the blanking variant achieving lower energy consumption than the non-blanking variant.

Takeaway 1: EAPO-NET converges reliably under both blanking and non-blanking configurations while learning energy-efficient policies.

C. Analysis of Energy Consumption

1) *Energy consumption as a function of load rate:* We evaluate the impact of the load rate with the average number of UEs fixed to 10. Fig. 2 reports the received load per UE, energy consumption per bps, and per-slot decision time for both RB-blanking and non-blanking.

Fig. 2(left) shows that energy consumption increases with the load rate. At RU saturation (2.5 and 5 Mbps), blanking opportunities disappear, and both curves overlap until additional RUs are activated. At high loads (6.5–8.25 Mbps), blanking reduces energy consumption by 10–20% with EAPO-NET, close to B&C (11–20%).

Fig. 2(center) shows that energy per bps decreases with the load rate, indicating better RB utilization. Fig. 2(right) shows that EAPO-NET requires only 1–7 ms (blanking) and 1–5 ms

(non-blanking), whereas B&C requires 0.35–165 s and 0.17–149 s, respectively.

Takeaway 2: EAPO-NET achieves near-optimal performance with an optimality gap of 0.7–3% (blanking) and 0.8–2.2% (non-blanking), while reducing decision time from seconds to milliseconds. Increasing the load rate has only a limited impact on latency.

2) *Energy consumption as a function of UEs density:* For a fixed average load of 1.5 Mbps per UE, Fig. 3 reports the energy consumption, energy consumption per UE, and per-slot decision time as the number of UEs increases.

Fig. 3(left) shows that energy consumption increases with the number of UEs. At RU saturation points (15, 30, 48, and 65 UEs), blanking opportunities disappear until additional RUs are activated. The largest energy savings are achieved between 33 and 48 UEs, reaching 18.4% with EAPO-NET and 18.3% with the optimal model.

Fig. 3(center) shows that energy consumption per UE decreases with increasing UE density due to improved RB utilization and amortized static energy costs. Fig. 3(right) shows that EAPO-NET requires only 1.9–13 ms (blanking) and 1.7–12.3 ms (non-blanking), whereas B&C requires 0.35–999 s and 0.3–856 s, respectively.

Takeaway 3: EAPO-NET maintains near-optimal performance while reducing decision time from seconds to milliseconds, with its computational advantage increasing as the number of UEs grows.

D. Robustness and Scalability Evaluation

We evaluate the robustness and scalability of EAPO-NET in a larger network by fixing the average number of UEs to 20 and varying the per-UE load from 6 to 12 Mbps.

Fig. 4(left) shows that EAPO-NET closely follows the B&C optimum, with an optimality gap of 0.1–0.5% (blanking) and 0.4–1.25% (non-blanking). The gap increases slightly above 10.5 Mbps as RB constraints become tighter, while remaining close to the optimum. Fig. 4(center) shows that energy per bit decreases with load for both approaches, whereas the benefit of blanking gradually diminishes as RB availability becomes limited.

Fig. 4(right) shows that the decision time of EAPO-NET remains nearly constant across all loads, while B&C grows rapidly with problem complexity, reaching 999 s per time slot at 12 Mbps.

Takeaway 4: EAPO-NET maintains near-optimal performance under high traffic while providing millisecond-scale decisions, demonstrating its scalability and suitability for real-time O-RAN control.

Problem-size scalability. We fix the average per-UE load at 2.5 Mbps (8 RBs per UE) and vary the number of UEs from 50 to 150, isolating the impact of problem size while keeping the per-UE resource demand constant.

Solution quality. Fig. 5(left) and Fig. 5(center) show the total and per-UE energy consumption. Across all network sizes, EAPO-NET closely follows the B&C optimum, while

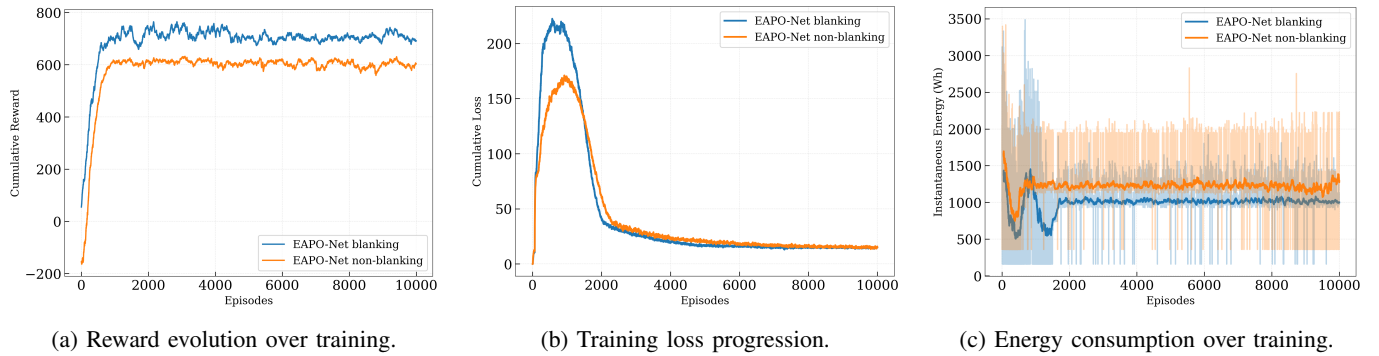


Fig. 1: Analysis of reward, loss, and energy consumption.

blanking consistently outperforms non-blanking. As the number of UEs increases, the optimality gap remains below 2.2%, demonstrating that EAPO-NET maintains its solution quality at larger scales.

Computational scalability. Fig. 5(right) shows that the B&C decision time grows rapidly with the number of UEs, whereas EAPO-NET remains nearly constant at 35–40 ms per slot. At 150 UEs, EAPO-NET is approximately 4×10^3 times faster than B&C.

Takeaway 5: EAPO-NET maintains a bounded optimality gap while providing nearly constant decision time, making it suitable for large-scale deployments where B&C becomes computationally prohibitive.

E. Sensitivity Analysis

We evaluate two key properties of EAPO-NET.

Fig. 6(a) shows the evolution of RB utilization during training. After an initial exploration phase, the policy converges by consolidating traffic onto a single active RU, sleeping the remaining RUs, and blanking nearly all idle RBs, resulting in negligible wasted RBs.

Fig. 6(b) shows the admission rate as a function of the energy-cost weight α in Eq. (12). The learned policy correctly prioritizes *priority 1* UEs, rejecting *priority 3* traffic first, followed by *priority 2*. The admission curves transition near the analytical break-even thresholds $\alpha_{\text{priority3}}^* = 1.94$, $\alpha_{\text{priority2}}^* = 2.78$, and $\alpha_{\text{priority1}}^* = 4.17$, validating the reward formulation. Even for large α , *priority 1* admission remains around 60%, indicating that the agent balances revenue and energy rather than optimizing either objective alone.

Takeaway 6: The weighted reward enables energy-aware and priority-aware admission without handcrafted rules. The analytical threshold α^* provides a practical guideline for tuning the energy-revenue trade-off.

VI. RELATED WORK

Mobile networks face a critical challenge: “energy consumption” is rapidly increasing while traffic becomes more dynamic and unpredictable. Current O-RAN control strategies are largely *static or rule-based*, keeping RAN and computing resources active during low traffic and wasting energy. Existing approaches mainly rely on cell sleeping (Cell Switching ON/OFF) and workload consolidation.

A large body of work exploits traffic fluctuations to deactivate base stations or sectors. For example, [8], [25] jointly optimize cell clustering and BS sleeping to consolidate load during medium- and long-term low-traffic periods. However, these approaches require strong inter-cell coordination, which is difficult to achieve in dense and heterogeneous deployments.

At shorter timescales, energy savings can be achieved without shutting down entire cells. Cell Discontinuous Transmission (DTX) [6], [7] enables sub-TTI micro-sleep during idle periods, while recent O-RAN xApps [26]–[29] exploit Advanced Sleep Modes (ASMs) and energy-aware scheduling. However, they generally assume ideal cell overlap or seamless UE mobility, limiting their applicability. Other works optimize UE–RU association, RB allocation, beamforming, or VNF placement [14], [19], [20], [30]–[33]. While effective for resource management and orchestration, they rarely exploit “RB blanking”, which reduces energy consumption within active RUs without requiring cell consolidation or ASM activation, and remains largely unexplored in O-RAN [34], [35].

Building on these advances, EAPO-NET makes three contributions: (i) a joint optimization framework for RU sleep control (ON/OFF and RB blanking), UE admission, and DU/CU VNF placement; (ii) a distributed multi-agent architecture with RU xApps, an admission xApp, and VNF dApps trained under CTDE with VDN and executed independently; and (iii) a comprehensive evaluation against exact ILP solutions to demonstrate scalability, adaptability, and solution quality.

VII. CONCLUSION

This paper presented EAPO-NET, a multi-agent DRL framework for the joint optimization of RU sleep mode, UE admission control, and DU/CU VNF placement in 5G O-RAN systems. By decomposing the problem across heterogeneous agents and leveraging CTDE with VDN, EAPO-NET enables coordinated learning while supporting fully decentralized execution without inter-agent communication. Extensive simulations demonstrated that EAPO-NET learns near-optimal policies that effectively balance energy consumption, network profit, and SLA compliance. Future work will extend EAPO-NET to multi-cell scenarios and investigate online policy adaptation mechanisms to address long-term traffic dynamics.

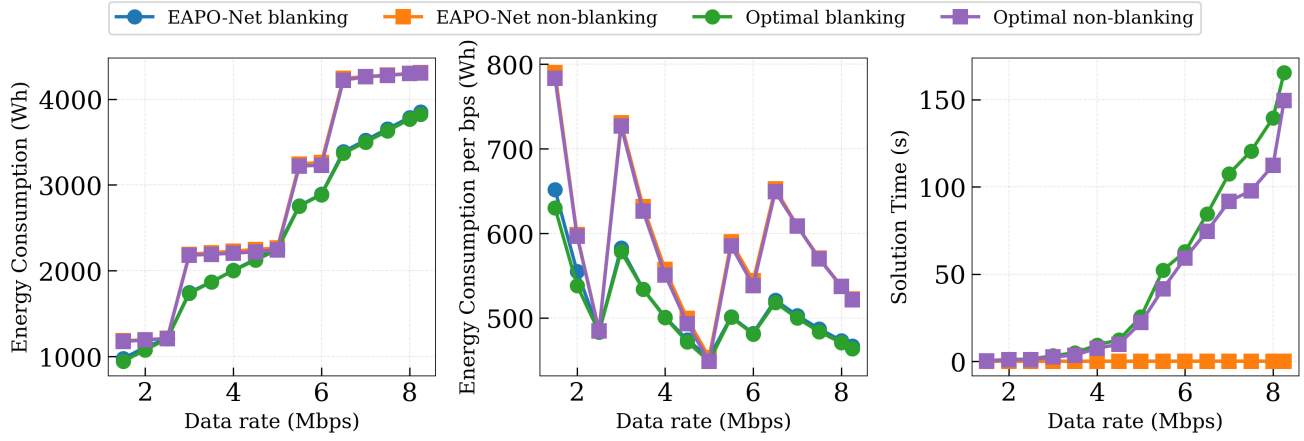


Fig. 2: Impact of load on energy consumption (left), energy per bps (center), and solution time (right).

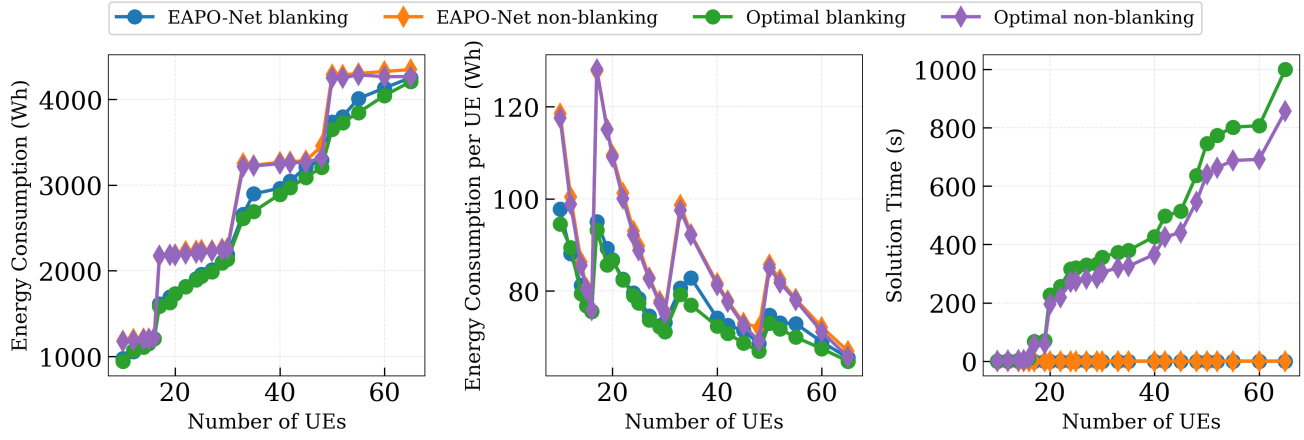


Fig. 3: Impact of UE count on energy consumption (left), energy per UE (center), and solution time (right).

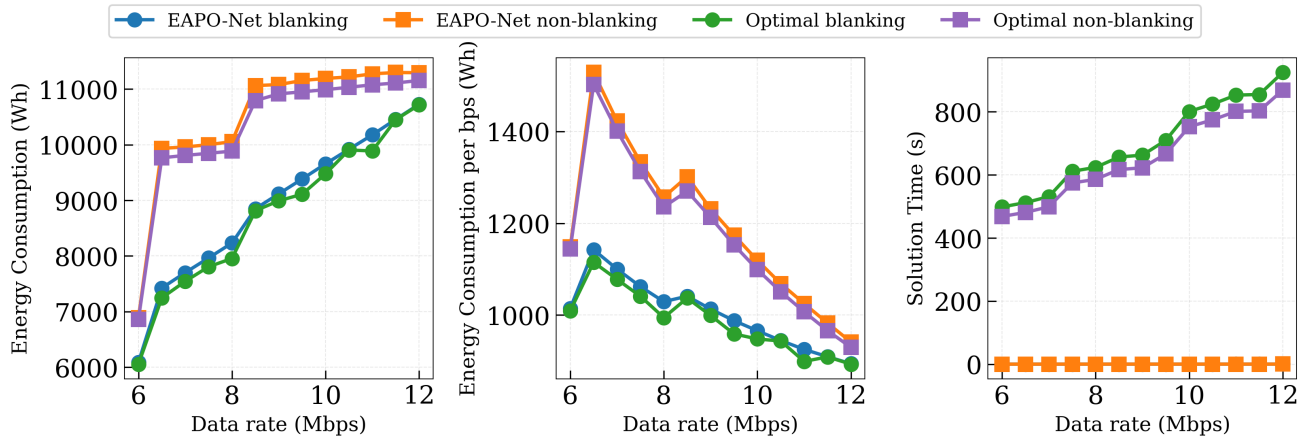


Fig. 4: Impact of load on energy consumption (left), energy per bps (center), and solution time (right).

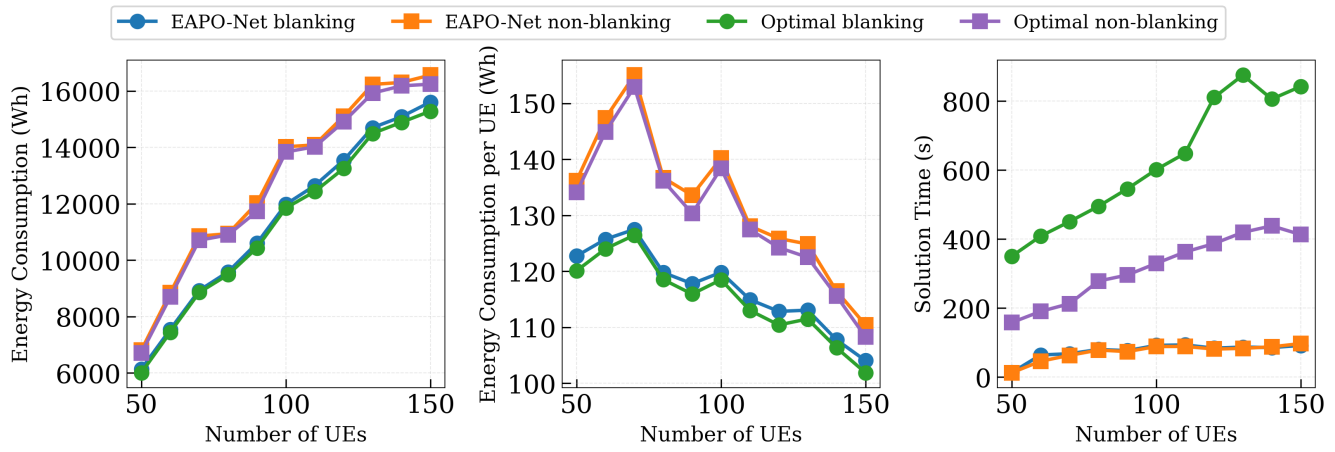
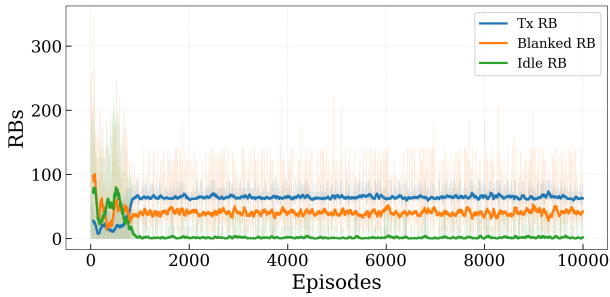
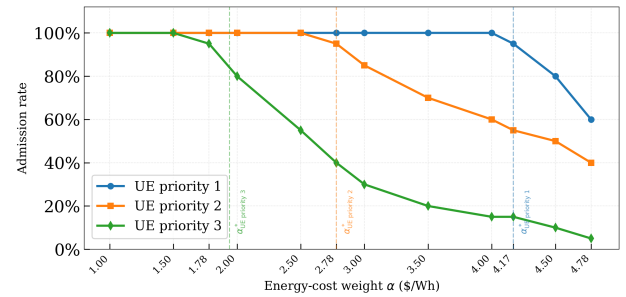


Fig. 5: Impact of UE count on energy consumption (left), energy per UE (center), and solution time (right).



(a) Evolution of Tx RBs, Blanking RBs, and Idle RBs through-out execution.



(b) Admission rate as a function of energy-cost weight α (\$/Wh).

Fig. 6: Admission rate versus the energy-cost weight α . Dashed lines denote the break-even thresholds $\alpha^* = H(y)/E(x, y, z)$, showing that lower-priority UEs are rejected first.

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